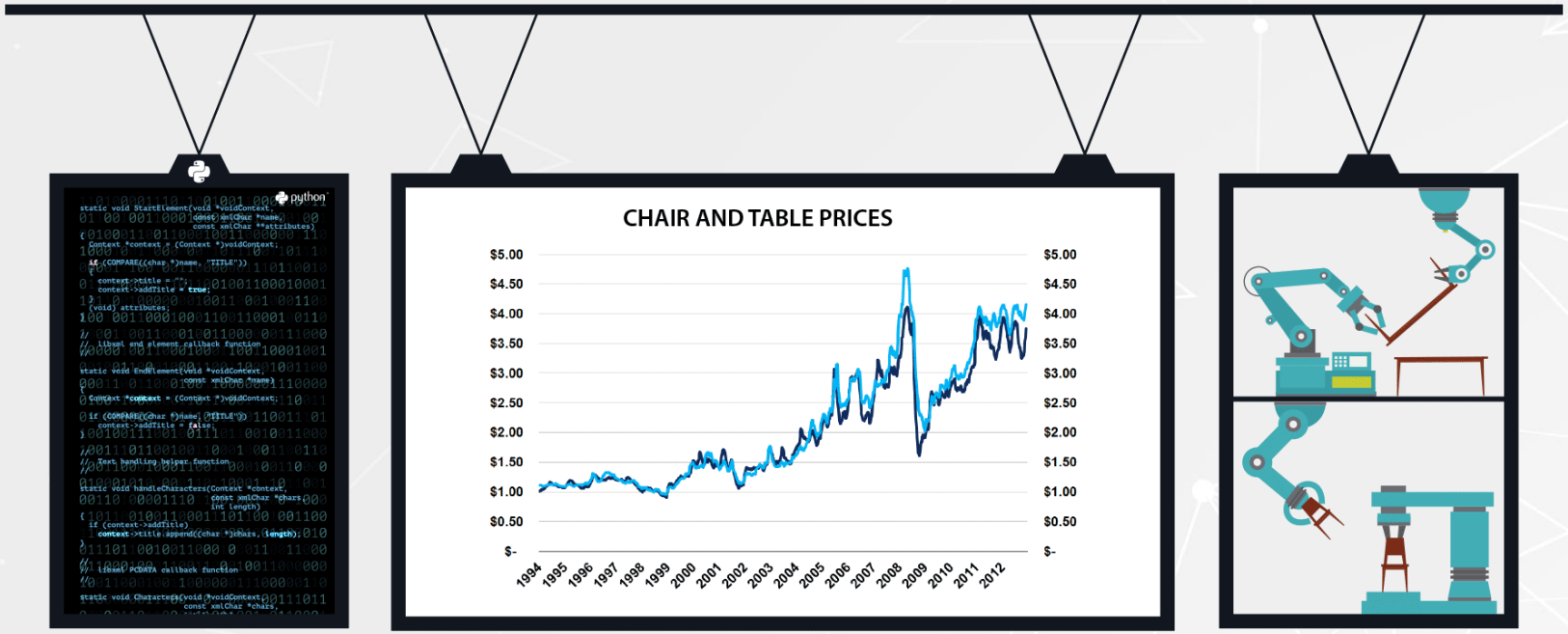
Multiple optimal solutions

Gurobi Python API

GUROBI OPTIMIZATION

Furniture Problem



New business conditions

- The data scientist in charge of production planning for the furniture factory uses machine learning to predict that the market price of a chair is now $50 and the market price of a table is $75.
- The equation of the objective function under these new business conditions is: revenue = 50x1 + 75x2.

Furniture Problem

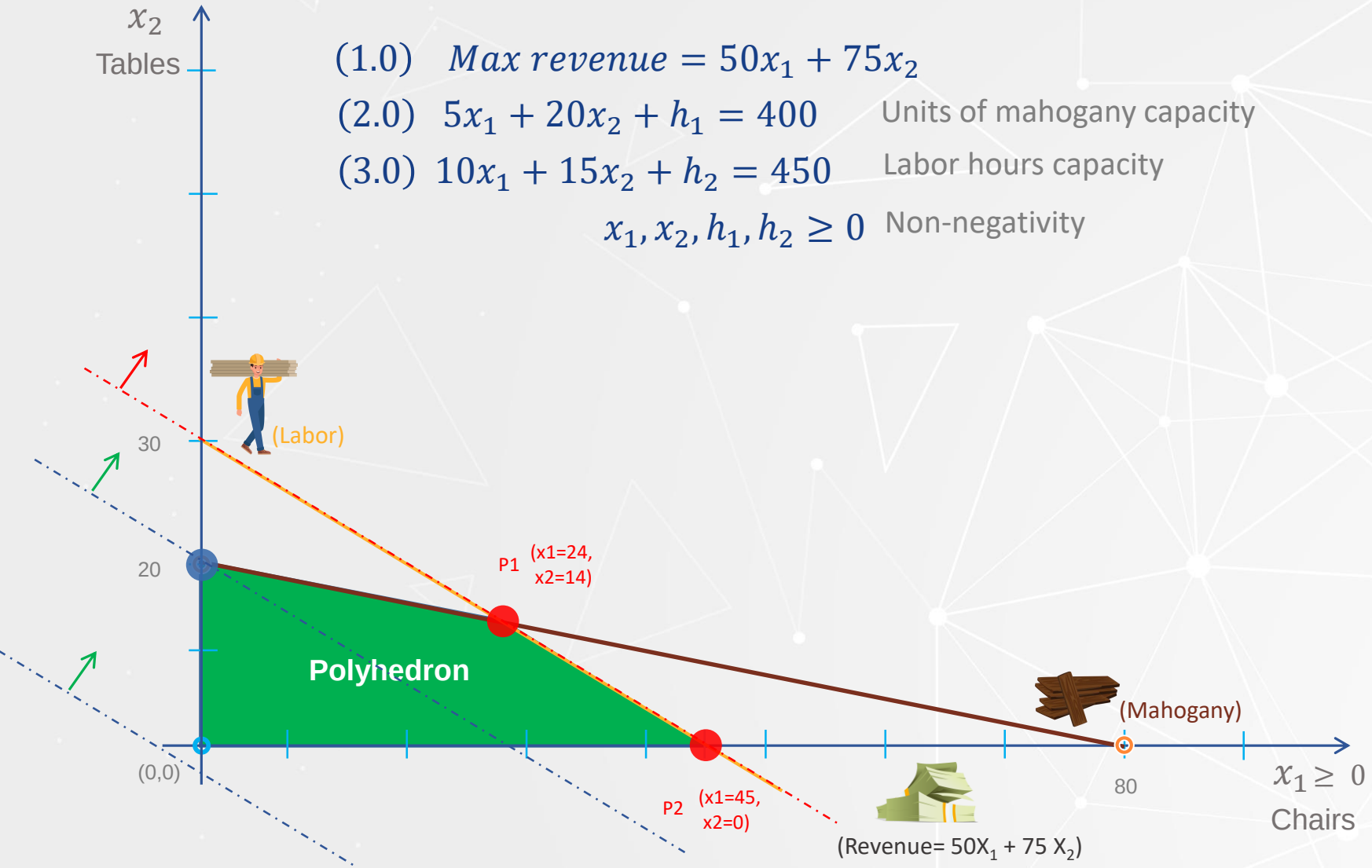
Furniture problem standard form

$$(1.0)\quad Max\ revenue = 50x_1 + 75x_2$$

$$(2.0)\quad 5x_1 + 20x_2 + h_1 = 400 \qquad \text{Units of mahogany capacity}$$

$$(3.0)\quad 10x_1 + 15x_2 + h_2 = 450 \qquad \text{Labor hours capacity}$$

$$x_1, x_2, h_1, h_2 \geq 0 \qquad \text{Non-negativity}$$



New business conditions

- In this case, the production plan P1 of building 24 chairs and 14 tables and the plan P2 of building 45 chairs and 0 tables are both optimal.
- The production plan P1 is defined by the mahogany and labor constraints. The associated optimal basic feasible solution is x1=24, x2=14, h1=0, h2=0.
- The production plan P2 is defined by the labor and a non-negativity constraint (x2 = 0), i.e. no tables are built. The associated optimal basic feasible solution is x1=45, x2=0, h1=175, h2=0.

Modified Furniture Problem: solved with Gurobi … 1

General Furniture model formulation -alternative optimal solutions

Let $price_p$ be the price of product $p \in products = \{chairs,\ tables\}$, and let $capacity_r$ be the capacity available of resource $r \in resources = \{mahogany,\ labor\}$.

Let $bom_{r,j}$ be the amount of resource r required by product p. Then the general formulation of the Furniture problem is:

Parametrized furniture LP problem formulation

$$Max \sum_{p \in products} price_p make_p$$

$$Subject\ to:$$

$$\sum_{p \in products} bom_{r,p} make_p \leq capacity_r \quad \forall\, r \in resources$$

$$make_p \geq 0 \quad \forall\, p \in products$$

We modify the objective function as follows:

New	Price
Chairs	$50
Tables	$75

```
# products data.
products, price = multidict({
  'chair': 50,
  'table':  75 })
```

New data, same model formulation

Modified Furniture Problem: solved with Gurobi … 2

```
# run optimization engine
f.optimize()

# display optimal values of decision variables
for v in f.getVars():
    print(v.varName, v.x)

# display optimal total profit value
print('total revenue', f.objVal)
```

```
('make[table]', 0.0)
('make[chair]', 45.0)
('total revenue', 2250.0)
```

Gurobi found one of the optimal solutions ... P2

```
# display shadow prices of resources constraints
for r in res:
    print(res[r].ConstrName, res[r].Pi)
```

```
('R[mahogany]', 0.0)
('R[labor]', 5.0)
```

Notice that the shadow price of the resource mahogany for this alternative optimal solution is zero,
… the marginal value of mahogany for this optimal solution is zero.

```
# display reduced costs of decision variables
for v in f.getVars():
    print(v.varName, abs(v.rc))
```

```
('make[table]', 0.0)
('make[chair]', 0.0)
```

Reduced cost of the non basic variable (tables) is zero. This means that if we produce more tables (and produce less chairs) the revenue generated remains the same.

Furniture Problem: simplex method revisited .. 1

New (original) LP problem

$$Max\ z = 50x1 + 75x2$$
$$s.t. 5x1 + 20x2 \leq 400$$
$$10x1 + 15x2 \leq 450$$
$$x1, x2 \geq 0$$

LP problem in canonical form with respect to the optimal basic variables (x1, h1) found by Gurobi:

$$Max\ z = 2{,}250 + (0x1 + 0h1) + (0x2 - 5h2)$$

Non basic variables

$$h1 = 175 - \left({}^{25}/_{2}\right)x2 + \left({}^{1}/_{2}\right)h2 \quad (2.0)$$
$$x1 = 45 - \left({}^{3}/_{2}\right)x2 - \left({}^{1}/_{10}\right)h2 \quad (3.0)$$
$$x1, x2, h1, h2 \geq 0$$

The LP problem in a standard form is

$$Max\ z = 50x1 + 75x2 + 0h1 + 0h2$$

$$s.t. 5x1 + 20x2 + h1 = 400$$
$$10x1 + 15x2 + h2 = 450$$
$$x1, x2, h1, h2 \geq 0$$

The reduced cost of non basic variable x2 is zero, hence if we increase its value, the optimal objective function value does not change. Let's decide which basic variable should become non basic (value of zero) by computing the minimum ratio test: $min\{175/(25/2) = 14, 45/(3/2) = 30\} = 14$; therefore h1 will become non basic variable. Hence, we pivot on constraint (2.0) and the column of variable x2.

Furniture Problem: simplex method revisited .. 2

LP problem in canonical form with respect to the basic variables (x1, x2):

$$Max\ z = 2{,}250 + 0h1 - 5h2$$

$$x2 = 14 - \left({}^{2}/_{25}\right)h1 + \left({}^{1}/_{25}\right)h2 \quad (2.0)$$

$$x1 = 24 - \left({}^{3}/_{25}\right)h1 - \left({}^{4}/_{25}\right)h2 \quad (3.0)$$

$$x1, x2, h1, h2 \geq 0$$

Modeling Opportunity

From the mathematical model point of view, we have two alternative optimal solutions S1= (x1 =24, x2 =14, h1=0, h2= 0) and S2 = (x1 =45, x2 =0, h1= 175, h2 = 0), both with a maximum total revenue of $2,250.

The data scientist points out that from the business perspective, the “optimal” solution S1 is preferred to the “optimal” solution S2, because the latter solution wastes 175 units of mahogany.

The data scientist decides to modify the LP model by now interpreting the slack variables as the amount of wasted resources, and defines the new decision variables:
x3 is the amount of unused mahogany and x4 is the amount of unused labor.

The data scientists uses machine learning to predict that the per unit inventory carrying cost of mahogany is $1 , and the unused per hour labor cost is $2. Then, the new LP model formulation is:

Furniture Problem: simplex method revisited .. 2

Modeling Opportunity
From the mathematical model point of view, we have two alternative optimal solutions S1= (x1 =24, x2 =14, x3 = x4 = 0) and S2 = (x1 =45, x2 =0, x3 = 175, x4 = 0), both with a maximum total revenue of $2,250.

The data scientist notice that from the business perspective, the “optimal” solution S1 is preferred to “optimal” solution S2, because with the latter solution is wasting 175 units of mahogany.

The data scientist decides to modify the LP model by now interpreting the slack variables as the amount of wasted resources, and defines the new decision variables:
x3 is the amount of unused mahogany and x4 is the amount of unused labor.

The data scientists using machine learning estimates that the per unit inventory carrying cost of mahogany is $1 , and the unused per hour labor cost is $2. Then, the new LP model formulation is:

$$Max\ z = 50x1 + 75x2 - 1x3 - 2x4$$

$$s.t.\ 5x1 + 20x2 + x3 = 400$$
$$10x1 + 15x2 + x4 = 450$$
$$x1, x2, x3, x4 \geq 0$$

New Furniture Problem: solved with Gurobi … 1

General Furniture model formulation -Penalize wasting resources

Let $price_p$ be the price of product $p \in products = \{chairs,\ tables\}$, and let $capacity_r$ be the capacity available of resource $r \in resource = \{mahogany,\ labor\}$.

Let $waste_r$ be a new decision variable that measures the amount of unused resource $r \in resoruces = \{mahogany,\ labor\}$. Let $cost_r$ be the per unit cost of unused resource capacity.

Let $bom_{r,j}$ be the amount of resource r required by product p. Then the new formulation of the Furniture problem is:

$$Max \sum_{p \in products} price_p make_p - \sum_{r \in resources} cost_r waste_r$$

$$Subject\ to:$$

$$\sum_{p \in products} bom_{r,p} make_p + waste_r = capacity_r \quad \forall\ r \in resources$$

$$make_p \geq 0 \quad \forall\ p \in products$$

$$waste_r \geq 0 \quad \forall\ r \in resources$$

Price

Per unit	Price
Chairs	$50
Tables	$75

Resources cost table:

Per unit	Cost
Mahogany	$1
Labor	$2

New Furniture Problem: solved with Gurobi … 2

```
# import gurobi library
from gurobipy import *
```

```
# resources data
resources, capacity, cost = multidict({
  'mahogany': [400, 1],
  'labor':  [450, 2] })
```

We add a cost parameter to the resources multidict to penalize the waste of resources.

```
# products data,
products,  price = multidict({
  'chair': 50,
  'table':  75 })
```

```
# Bill of materials: resources required by each product
bom = {
  ('mahogany', 'chair'): 5,
  ('mahogany', 'table'):  20,
  ('labor', 'chair'):      10,
  ('labor', 'table'):   15 }
```

New Furniture Problem: solved with Gurobi … 3

```
# Declare and initialize model
f = Model('Furniture')
```

```
# Create decision variables for the products to make
make = f.addVars(products, name="make")
```

```
# Create decision variables for the wasted resources
waste = f.addVars(resources, name="waste")
```

We add a new type of decision variable to measure the unused resource capacity.

```
# Capacity resource constraints:
# The amount consumed by the products made + the waste = capacity

res = f.addConstrs(((sum(bom[r,p]*make[p] for p in products) + waste[r] == capacity[r]) for r in resources), name='R')
```

$$\sum_{p \in products} bom_{r,p} make_p + waste_r = capacity_r \quad \forall\, r \in resources$$

```
# The objective is to maximize total profit
f.setObjective(make.prod( price ) - waste.prod(cost) , GRB.MAXIMIZE)
```

$$Max \sum_{p \in products} price_p make_p - \sum_{r \in resources} cost_r waste_r$$

New Furniture Problem: solved with Gurobi … 4

```
# run optimization engine
f.optimize()
```

```
# display optimal values of decision variables
for v in f.getVars():
    print(v.varName, v.x)

# display optimal total profit value
print('total profits', f.objVal)
```

```
('make[table]', 14.0)
('make[chair]', 24.0)
('waste[mahogany]', 0.0)
('waste[labor]', 0.0)
('total profits', 2250.0)
```

The optimal solution is now making 14 tables and 24 chairs with a total objective function value of $2,250.

Notice that the solution of making 0 tables and 45 chairs is no longer optimal since the objective function value is
($50*45 = $2,250) – ($1*175 = $175) = $2,075.
That is, the alternative solution (chairs = 45, tables =0) is no longer optimal.